

SMART CRIME ANALYSIS AND CRIMINAL IDENTIFICATION SYSTEM

Problem Statement

Develop an intelligent crime policing system that helps police departments analyze crime data, identify criminals, and predict possible suspects using AI, crime patterns, facial recognition, and location-based analysis.

The system should:

- Store and manage criminal records securely.
 - Analyze previous crime data to identify crime hotspots.
 - Predict possible criminals based on crime style (modus operandi).
 - Match facial details with criminal databases using face recognition.
 - Suggest possible suspects from a particular area where the crime occurred.
 - Visualize crime statistics and trends for police investigation support.
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Main Objectives

1. Identify criminals using facial recognition.
 2. Predict possible suspects based on previous crime methods.
 3. Analyze crime-prone areas using crime data visualization.
 4. Maintain complete criminal records digitally.
 5. Help police reduce investigation time using AI-based suggestions.
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Modules of the Project

1. Criminal Record Management

Stores:

- Criminal name
- Age
- Address
- Crime history
- Fingerprints
- Facial images
- Gang information
- Previous arrest records

Output:

- Digital criminal profiles

- Searchable criminal database
 - Complete history reports
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2. Face Recognition Criminal Detection

Uses CCTV images or uploaded photos to identify criminals.

Output:

- Face match percentage
 - Suspect identity
 - Criminal record details
 - Alert if criminal found
-

3. Crime Pattern Analysis

Analyzes:

- How the crime was committed
- Weapon used
- Entry method
- Time of crime
- Crime type

Matches it with past criminal behavior patterns.

Output:

- Possible suspect list
 - Crime style matching report
 - Similar previous crime cases
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4. Area-wise Criminal Prediction

If crime occurs in a particular location, the system checks:

- Criminals living nearby
- Previously active criminals in that area
- Similar crime history in that location

Output:

- Area-based suspect list
 - Nearby criminal records
 - Risk-level analysis map
-

5. Crime Data Visualization Dashboard

Visual representation of:

- Crime hotspots
- Monthly/yearly crime rates
- Crime category distribution
- Dangerous areas

Output:

- Heatmaps
 - Pie charts
 - Bar graphs
 - Interactive crime dashboard
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6. AI-Based Crime Prediction

Machine learning model predicts:

- Probability of future crimes
- Most likely crime locations
- Possible repeat offenders

Output:

- Crime prediction reports
 - Future crime trend graphs
 - High-risk area alerts
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Technologies That Can Be Used

Purpose	Technologies
Frontend	HTML, CSS, JavaScript, React
Backend	Python Flask / Django
Database	MySQL / MongoDB
AI/ML	Scikit-learn, TensorFlow
Face Recognition	OpenCV, face_recognition library
Visualization	Matplotlib, Plotly, Chart.js

Expected Final Output of the Project

The system will provide:

- Criminal identification using face recognition
- Suspect prediction based on crime behavior
- Crime hotspot visualization

- Area-wise criminal analysis
- Digital criminal database
- AI-generated suspect recommendations
- Investigation support dashboard

Complete Dataset Structure for Crime Analysis and Criminal Identification System

1. Crime Records Dataset

Table Name: `crime_records`

Column Name	Data Type	Description
crime_id	INT / VARCHAR	Unique crime ID
crime_type	VARCHAR	Theft, Murder, Robbery
crime_subtype	VARCHAR	Chain Snatching, Cyber Fraud
crime_date	DATE	Crime date
crime_time	TIME	Crime time
crime_day	VARCHAR	Monday, Tuesday
crime_month	VARCHAR	January
crime_year	INT	2026
crime_location	VARCHAR	Full location
area_name	VARCHAR	Chinchwad
city	VARCHAR	Pune
state	VARCHAR	Maharashtra
pincode	INT	Postal code
latitude	FLOAT	GPS latitude
longitude	FLOAT	GPS longitude
crime_description	TEXT	Detailed description
weapon_used	VARCHAR	Knife, Gun
entry_method	VARCHAR	Door break, Window
crime_method	TEXT	How crime was committed
target_type	VARCHAR	House, Bank, Shop
victim_count	INT	Number of victims
victim_gender	VARCHAR	Male/Female
victim_age	INT	Victim age
injury_level	VARCHAR	Minor/Severe
property_loss_amount	FLOAT	Money/property loss
suspect_count	INT	Number of suspects
cctv_available	BOOLEAN	CCTV available or not
evidence_found	TEXT	Fingerprint, Weapon
police_station	VARCHAR	Police station name
officer_assigned	VARCHAR	Officer name
case_status	VARCHAR	Solved/Unsolved
fir_number	VARCHAR	FIR number
arrest_made	BOOLEAN	Arrest status
criminal_id	VARCHAR	Linked criminal ID

investigation_notes	TEXT	Police notes
created_at	DATETIME	Record creation time

2. Criminal Database Dataset

Table Name: `criminal_records`

Column Name	Data Type	Description
criminal_id	VARCHAR	Unique criminal ID
first_name	VARCHAR	First name
last_name	VARCHAR	Last name
full_name	VARCHAR	Full name
nickname	VARCHAR	Criminal alias
gender	VARCHAR	Male/Female
age	INT	Age
dob	DATE	Date of birth
blood_group	VARCHAR	Blood group
nationality	VARCHAR	Nationality
religion	VARCHAR	Religion
marital_status	VARCHAR	Married/Single
phone_number	VARCHAR	Contact number
email	VARCHAR	Email
aadhaar_number	VARCHAR	Government ID
passport_number	VARCHAR	Passport ID
address	TEXT	Full address
area_name	VARCHAR	Residential area
city	VARCHAR	City
state	VARCHAR	State
pincode	INT	Postal code
latitude	FLOAT	Location latitude
longitude	FLOAT	Location longitude
gang_name	VARCHAR	Gang affiliation
gang_role	VARCHAR	Gang role
education_level	VARCHAR	Qualification
occupation	VARCHAR	Job
criminal_category	VARCHAR	Violent/Non-violent
risk_level	VARCHAR	Low/Medium/High
fingerprint_id	VARCHAR	Fingerprint record
dna_profile_id	VARCHAR	DNA profile
eye_color	VARCHAR	Eye color
hair_color	VARCHAR	Hair color
height_cm	FLOAT	Height
weight_kg	FLOAT	Weight
body_marks	TEXT	Tattoos/scars
face_image_path	VARCHAR	Face image location
mugshot_image	VARCHAR	Mugshot path
last_known_location	VARCHAR	Last location
last_seen_date	DATE	Last seen
arrest_count	INT	Total arrests
conviction_count	INT	Convictions
prison_history	TEXT	Jail history

bail_status	VARCHAR	On bail or not
most_common_crime	VARCHAR	Frequent crime
crime_style	TEXT	Modus operandi
preferred_weapon	VARCHAR	Common weapon
active_status	BOOLEAN	Active criminal
wanted_status	BOOLEAN	Wanted or not
created_at	DATETIME	Record creation

3. Crime Modus Operandi Dataset

Table Name: crime_modus_operandi

Column Name	Data Type	Description
mo_id	INT	MO record ID
criminal_id	VARCHAR	Criminal ID
crime_type	VARCHAR	Robbery, Theft
entry_method	VARCHAR	Window break
escape_method	VARCHAR	Bike escape
target_selection	VARCHAR	Jewelry shops
crime_timing	VARCHAR	Midnight
weapon_used	VARCHAR	Knife
dress_pattern	VARCHAR	Black hoodie
communication_method	VARCHAR	Phone/Signal
vehicle_used	VARCHAR	Bike/Car
accomplice_count	INT	Number of partners
victim_selection	VARCHAR	Elderly people
planning_level	VARCHAR	Planned/Random
technology_used	VARCHAR	GPS, Hacking
repeat_pattern	TEXT	Repeated behaviors
special_notes	TEXT	Additional info

4. Face Recognition Dataset

Table Name: face_recognition_data

Column Name	Data Type	Description
face_id	INT	Face record ID
criminal_id	VARCHAR	Linked criminal
image_path	VARCHAR	Image location
face_encoding	BLOB	Face vector encoding
image_resolution	VARCHAR	720x720
capture_angle	VARCHAR	Front/Side
lighting_condition	VARCHAR	Bright/Low light
facial_expression	VARCHAR	Neutral/Angry
beard_status	BOOLEAN	Beard present
glasses_status	BOOLEAN	Wearing glasses
mask_status	BOOLEAN	Wearing mask
capture_date	DATETIME	Image date
source_type	VARCHAR	CCTV/Upload

confidence_score	FLOAT	Match confidence
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5. Area Crime Statistics Dataset

Table Name: area_crime_statistics

Column Name	Data Type	Description
area_id	INT	Area ID
area_name	VARCHAR	Area name
city	VARCHAR	City
state	VARCHAR	State
latitude	FLOAT	Latitude
longitude	FLOAT	Longitude
total_crimes	INT	Total crimes
murder_count	INT	Murder cases
robbery_count	INT	Robbery cases
theft_count	INT	Theft cases
cybercrime_count	INT	Cyber crimes
women_crime_count	INT	Crimes against women
juvenile_crime_count	INT	Juvenile crimes
night_crime_count	INT	Night crimes
day_crime_count	INT	Day crimes
repeat_offender_count	INT	Repeat criminals
hotspot_score	FLOAT	Crime risk score
most_common_crime	VARCHAR	Frequent crime
safest_time	VARCHAR	Low-risk timing
dangerous_time	VARCHAR	High-risk timing
last_updated	DATETIME	Last update

6. CCTV Surveillance Dataset

Table Name: cctv_surveillance

Column Name	Data Type	Description
footage_id	INT	Video ID
camera_id	VARCHAR	Camera ID
camera_location	VARCHAR	Camera location
latitude	FLOAT	Latitude
longitude	FLOAT	Longitude
recording_date	DATE	Recording date
recording_time	TIME	Recording time
video_path	VARCHAR	Video file
detected_faces	INT	Faces detected
suspicious_activity	BOOLEAN	Suspicious activity
matched_criminal_id	VARCHAR	Matched criminal
match_confidence	FLOAT	Confidence score
object_detected	VARCHAR	Weapon/Vehicle
alert_generated	BOOLEAN	Alert status

7. Police Officers Dataset

Table Name: `police_officers`

Column Name	Data Type	Description
<code>officer_id</code>	VARCHAR	Officer ID
<code>officer_name</code>	VARCHAR	Name
<code>rank</code>	VARCHAR	Inspector, SI
<code>badge_number</code>	VARCHAR	Badge number
<code>phone_number</code>	VARCHAR	Contact
<code>email</code>	VARCHAR	Email
<code>police_station</code>	VARCHAR	Station name
<code>assigned_area</code>	VARCHAR	Area
<code>solved_cases</code>	INT	Solved case count
<code>active_cases</code>	INT	Ongoing cases
<code>specialization</code>	VARCHAR	Cybercrime/Homicide
<code>duty_status</code>	VARCHAR	Active/Off duty

Best Database Choice

For this project:

- MySQL/PostgreSQL → Structured data
- MongoDB → Flexible crime reports
- Firebase → Real-time alerts
- Vector Database → Face embeddings (optional)

Recommended Dataset Formats

Module	Best Format
Crime Records	MySQL / CSV
Criminal Profiles	MySQL / JSON
Face Recognition	Image folders + CSV/JSON
Crime Prediction ML	CSV
CCTV Data	Video/Image files
Real-time Alerts	Firebase JSON
Geographic Data	GeoJSON

1. MySQL Format (Recommended Main Database)

Best for:

- Criminal records
- FIR records

- Police data
- Crime history

Example Table Format

```
CREATE TABLE criminal_records (  
    criminal_id VARCHAR(20),  
    full_name VARCHAR(100),  
    age INT,  
    crime_type VARCHAR(50),  
    area_name VARCHAR(50),  
    risk_level VARCHAR(20)  
);
```

Example Data

criminal_id	full_name	age	crime_type	area_name	risk_level
C101	Rahul Patil	32	Robbery	Chinchwad	High

Advantages

- Fast searching
 - Easy filtering
 - Secure
 - Good for large systems
-

2. CSV Format (Best for Machine Learning)

Best for:

- Crime prediction
- Data analysis
- Visualization

Example CSV

```
crime_id,crime_type,area_name,weapon_used,crime_time,risk_level  
CR101,Robbery,Pimpri,Knife,Night,High  
CR102,Theft,Chinchwad,None,Day,Medium
```

Advantages

- Easy for Python/Pandas
 - Used in ML training
 - Simple to edit
-

3. JSON Format (Best for Flexible Data)

Best for:

- API responses
- Dynamic crime reports
- Firebase storage
- Web applications

Example JSON

```
{
  "criminal_id": "C101",
  "full_name": "Rahul Patil",
  "age": 32,
  "crimes": [
    {
      "crime_type": "Robbery",
      "year": 2024
    },
    {
      "crime_type": "Theft",
      "year": 2025
    }
  ],
  "risk_level": "High"
}
```

Advantages

- Flexible structure
- Easy API integration
- Works well with NoSQL databases

4. Image Dataset Format (For Face Recognition)

Best for:

- Face matching
- Criminal detection

Folder Structure

```
criminal_faces/
├── C101/
│   ├── img1.jpg
│   └── img2.jpg
├── C102/
│   └── img1.jpg
```

Optional CSV Mapping

```
criminal_id,image_path
C101,criminal_faces/C101/img1.jpg
C102,criminal_faces/C102/img1.jpg
```

5. GeoJSON Format (For Crime Maps)

Best for:

- Heatmaps
- Crime hotspot visualization

Example GeoJSON

```
{
  "type": "Feature",
  "properties": {
    "crime_type": "Robbery",
    "area": "Pimpri"
  },
  "geometry": {
    "type": "Point",
    "coordinates": [73.7997, 18.6298]
  }
}
```

Advantages

- Used in maps
 - Works with Leaflet & Google Maps
-

6. Firebase JSON Structure (Real-Time Alerts)

Best for:

- Live alerts
- Mobile notifications
- Emergency systems

Example Firebase Data

```
{
  "alerts": {
    "alert_1": {
      "crime_type": "Robbery",
      "location": "Pimpri",
      "time": "11:30 PM",
      "suspect_detected": true
    }
  }
}
```

Best Overall Architecture for Your Project

Recommended Combination

Purpose	Format
Main Database	MySQL
ML Training	CSV
Face Recognition	Images + CSV
APIs	JSON
Maps	GeoJSON
Real-Time Features	Firebase

Recommended Technologies

Purpose	Technology
Database	MySQL
Backend	Flask/Django
ML	Scikit-learn
Face Recognition	OpenCV
Visualization	Plotly/Chart.js
Maps	Leaflet.js
Real-time	Firebase

Final Recommended Structure

```
project/
├── database/
│   ├── crime_records.sql
│   └── criminal_records.sql
├── datasets/
│   ├── crimes.csv
│   └── criminals.csv
├── faces/
│   ├── C101/
│   └── C102/
├── geo/
│   └── crime_hotspots.geojson
├── firebase/
│   └── alerts.json
```

Dummy Dataset Creation for Project Prototype

For your project prototype, you will usually need to create synthetic (dummy) datasets because real police/criminal datasets are restricted and sensitive.

1. Crime Records Dataset

Where to Get Realistic Base Data

- Kaggle Crime Datasets
- Data.gov.in Crime Data
- Chicago Crime Dataset

How to Create Dummy Data

Use:

- Python Faker library
- ChatGPT-generated CSV
- Random crime generators

Generate Fields Like

- Crime type
- Area
- Time
- Weapon used
- FIR number

Python Libraries

```
from faker import Faker
import random
```

2. Criminal Records Dataset

You WON'T Get Real Criminal Data Publicly.

So create synthetic profiles.

How to Create

Use:

- Faker library
- Random image datasets
- AI-generated faces

Generate:

- Fake names
- Addresses
- Crime histories
- Gang names
- Arrest counts

Best Tool

- Faker Python Library
-

3. Face Recognition Dataset

Best Sources

Public Face Datasets

- LFW Dataset
- CelebA Dataset

For Fake Criminal Faces

Use:

- This Person Does Not Exist

You can:

- Download AI-generated faces
- Rename them as criminal IDs

Example

C101_face1.jpg
C101_face2.jpg

4. Crime Modus Operandi Dataset

This is mostly custom-created.

Create Using:

- News articles
- Crime documentaries
- Random combinations

Example

crime_type	entry_method	weapon
Robbery	Window Break	Knife
Theft	Fake Key	None

Generate Using

- Excel
- Python random module
- ChatGPT

5. Area Crime Statistics Dataset

Sources

- Google Maps
- OpenStreetMap
- Kaggle crime datasets

Create Dummy Data

You can manually assign:

- Crime counts
- Risk scores
- Hotspot ratings

Example

Area Crime Count

Pimpri 120

Nigdi 90

6. CCTV Dataset

Easy Way

Use:

- Sample CCTV videos from YouTube
- Surveillance datasets

Sources

- UCF Crime Dataset
- Roboflow Universe

Simple Alternative

Use:

- Mobile-recorded videos
 - Campus CCTV-like videos
-

7. Police Officer Dataset

Fully Dummy Dataset.

Generate using Faker.

Fields:

- Officer name
 - Badge number
 - Rank
 - Assigned station
-

BEST WAY TO CREATE LARGE DUMMY DATASETS

Method 1 — Python Faker Library (BEST)

Install

```
pip install faker pandas
```

Example Generator

```
from faker import Faker
import pandas as pd
import random

fake = Faker()

data = []

for i in range(100):
    data.append({
        "criminal_id": f"C{i}",
        "name": fake.name(),
        "address": fake.address(),
        "crime_type": random.choice(["Robbery", "Theft", "Fraud"]),
        "age": random.randint(18, 60)
```

```
}}
```

```
df = pd.DataFrame(data)  
df.to_csv("criminals.csv", index=False)
```

Method 2 — ChatGPT Generated CSV

You can ask:

- “Generate 1000 crime records CSV”
 - “Generate criminal dataset with Indian names”
-

Method 3 — Mockaroo (VERY GOOD)

Website

[Mockaroo](#)

Features

- Generate fake datasets
- Export CSV/JSON/SQL
- Custom columns
- Large datasets

You can generate:

- Names
 - Crimes
 - Locations
 - Dates
 - Phone numbers
-

Method 4 — AI Generated Images

Use:

- This Person Does Not Exist

For:

- Criminal face datasets
 - Face recognition testing
-

Recommended Dataset Creation Strategy

Dataset	Best Creation Method
Crime Records	Kaggle + Faker
Criminal Profiles	Faker + Mockaroo
Face Dataset	AI-generated faces
Crime Pattern	Custom CSV
Area Statistics	Manual + Kaggle
CCTV	YouTube/sample videos
Police Dataset	Faker

Recommended Final Dataset Size

Dataset	Suggested Size
Crime Records	10,000+
Criminal Profiles	2,000+
Face Images	5–10 per criminal
Area Stats	100+ areas
Crime Patterns	5,000+
CCTV Clips	50–100 videos

Best Overall Stack for Your Project

Component	Recommended
Database	MySQL
Dataset Creation	Faker + Mockaroo
Face Recognition	OpenCV
ML	Scikit-learn
Visualization	Plotly
Backend	Flask
Frontend	React/Tailwind